

# Test Plan for US Bangla Airlines Reservation & Portal System

**Document Type:** Software Test Plan

**Product:** US Bangla Airlines Reservation & Portal System

**Platform:** Web Application

**Prepared By:** Pranjal Chowdhury

**Test Period:** 31 August 2026 – 4 September 2026

**Browsers:** Google Chrome, Microsoft Edge

**Primary Test Environment:** US Bangla Airlines Web Portal / Front Office System

---

## 1. Introduction

The purpose of this test plan is to outline the approach, objectives, scope, resources, and testing activities for the **US Bangla Airlines Reservation & Portal System**. This document provides a structured framework for systematically assessing the functionality, usability, reliability, compatibility, and input validation of the web based airline reservation and customer portal system.

This system offers several key services to customers, including profile registration, login, flight booking, flight status checks, profile management, and customer support. Therefore, thorough testing is required to ensure the system functions correctly and provides a reliable, user friendly experience.

### 1.1 Web Application Overview

US Bangla Airlines' reservation and portal system is a web based application designed to provide air travel related services to customers. Its key functionalities include creating new profiles, logging into accounts, searching for flights, checking flight status, updating profile information, and contacting customer support.

The system is expected to function correctly on supported web browsers and appropriately handle valid, invalid, boundary, and unexpected user inputs.

### 1.2 Purpose

The primary purpose of this test plan is to verify the quality, functionality, reliability, and usability of the US Bangla Airlines web application.

The testing process will ensure that:

- Critical customer facing functions work according to requirements.

- Registration and profile management work correctly.
  - Login and password recovery function properly.
  - Flight search and passenger selection operate correctly.
  - Flight status searches return appropriate results.
  - Mandatory and format validations work correctly.
  - Invalid and malformed inputs are handled safely.
  - The application does not freeze or crash.
  - The interface and error messages are properly displayed.
  - The application behaves consistently in Chrome and Edge.
  - Previously fixed defects do not reappear during regression testing.
- 

## 2. Scope of Testing Approach

### 2.1 Scope of Testing

The testing approach for the US Bangla Airlines Reservation & Portal System will encompass a comprehensive evaluation of its major features, functionality, validation, usability, security related input handling, compatibility, and integration behavior.

The major testing areas include:

#### A. Create a New Profile

Testing will include:

- Registration page accessibility
- Mandatory and optional fields
- Login ID validation
- Duplicate Login ID validation
- Special characters and whitespace
- Maximum Login ID length
- Password complexity
- Password confirmation
- Email format validation
- Unusual email domains
- Birth date validation
- First name and surname validation
- Hyphens and apostrophes
- Phone number validation
- Newsletter subscription
- Successful account creation

## **B. Login**

Testing will include:

- Login page accessibility
- Valid and invalid credentials
- Empty mandatory fields
- Case sensitivity
- Special character credentials
- Forgot Password
- Password reset email
- Password show/hide functionality
- Email validation

## **C. Book a Flight**

Testing will include:

- Homepage accessibility
- Round trip search
- One way search
- Multi city search
- Departure and destination validation
- Date selection
- Passenger selection
- Adult/child restrictions
- Maldivian Adult passenger category
- Promo code validation
- Flight search results

## **D. Flight Status**

Testing will include:

- Flight Status page accessibility
- Search by route
- Search by flight number
- Departing/Arriving filters
- Date selection
- Invalid date handling
- Mandatory airport validation
- Mandatory flight number validation
- Date picker behavior
- UI alignment

## **E. Update Your Profile**

Testing will include:

- Profile update page accessibility
- Mobile and office phone modification
- Login ID validation
- Email validation
- Birth date validation
- Password update
- Confirm password matching
- Mandatory fields
- Phone validation
- My Account dropdown behavior

## **F. Help / Contact Us**

Testing will include:

- Contact Us page accessibility
- Valid form submission
- Name validation
- Email validation
- Contact number validation
- Subject validation
- Message validation
- Mandatory field validation
- Invalid email formats
- Non numeric phone input
- Special character input
- Submission error handling

---

## **2.2 Out of Scope**

The following areas are outside the current testing scope:

- Native Android/iOS application testing
- Load and stress testing
- Full penetration testing
- Payment gateway certification
- Actual airline operational systems
- Backend database performance testing
- Third party infrastructure testing
- Disaster recovery testing

- Production deployment testing
- Physical airport and airline operational processes

However, any defect related to security, integration, application crashes, freezes, or serious performance problems will be reported for further investigation.

### 3. Testing Methodology

The following testing techniques will be used during the testing process:

| Testing Type              | Purpose                                 |
|---------------------------|-----------------------------------------|
| Functional Testing        | Verify system functionality             |
| Positive Testing          | Verify valid inputs work correctly      |
| Negative Testing          | Verify invalid inputs are rejected      |
| Boundary Testing          | Verify minimum and maximum limits       |
| Validation Testing        | Verify mandatory and format rules       |
| UI/UX Testing             | Verify layout and usability             |
| Cross Browser Testing     | Verify Chrome and Edge compatibility    |
| Security/Input Validation | Verify safe handling of malformed input |
| Integration Testing       | Verify email/SMS related functions      |
| Regression Testing        | Verify defect fixes                     |
| Smoke Testing             | Verify critical functions after a build |

### 4. Test Environment

The testing will be performed in the following environment:

| Item           | Environment                                    |
|----------------|------------------------------------------------|
| Application    | US Bangla Airlines Reservation & Portal System |
| Platform       | Web Application                                |
| Browser 1      | Google Chrome                                  |
| Browser 2      | Microsoft Edge                                 |
| Testing Method | Manual Testing                                 |
| Network        | Internet connected environment                 |
| Test Data      | Valid, Invalid, Boundary and Malformed Data    |

| Item     | Environment                     |
|----------|---------------------------------|
| Email    | Test Email Accounts             |
| SMS      | Test Mobile Numbers             |
| Evidence | Screenshots and Defect Evidence |

---

## 5. Test Data

The test data will consist of valid, invalid, boundary, and security/input validation data.

### 5.1 Valid Data

- Login ID: Test account
- Password: Test account password
- Email: Test email account
- Phone: Test mobile number

### 5.2 Invalid Data

- Email: xyz
- Email: ?"@gmail.com
- Phone: abcaksjnhdk
- Date: 50/5/2025

### 5.3 Boundary Data

- 300+ character Login ID
- Maximum length phone number
- Very long name
- Minimum length email
- Long password
- Extremely long message/text

### 5.4 Security/Input Validation Data

Examples include:

- a11\$""
- xyz@""
- jonee"
- ---jonee
- aa aaaa

The test repository should not contain real customer credentials or personally identifiable information.

---

## 6. Entry Criteria

Testing can begin when:

- Required build is deployed.
  - Test environment is accessible.
  - Required modules are available.
  - Test accounts are available.
  - Test data is prepared.
  - Supported browser versions are available.
  - Basic smoke testing has passed.
  - Major environment issues have been resolved.
- 

## 7. Exit Criteria

Testing can be considered complete when:

- All planned test cases have been executed.
- Critical and High defects have been fixed or formally accepted.
- Regression testing has been completed.
- No unresolved application crash/freeze defects remain.
- No blocking defects remain in critical customer journeys.
- Test evidence has been captured.
- Final test execution report has been prepared.
- Business/Product stakeholders approve outstanding issues.

For a production release, the number of **critical defects must be zero**, and high severity defects affecting registration, login, booking, or other key customer transactions must be resolved or formally accepted.

---

## 8. Defect Severity

| Severity | Definition                                                             |
|----------|------------------------------------------------------------------------|
| Critical | Application crash, freeze, security risk, or core function unavailable |

| Severity | Definition                                                      |
|----------|-----------------------------------------------------------------|
| High     | Major functionality fails or incorrect business behavior occurs |
| Medium   | Function works incorrectly but workaround exists                |
| Low      | Minor usability or cosmetic issue                               |
| Trivial  | Minor spelling or negligible cosmetic issue                     |

## 9. Regression Testing

After the defect is fixed, the relevant test cases and regression scenarios will be re-run to ensure that the issue has been resolved and that there is no negative impact on existing functionality.

For example, after fixing, the following should be retested:

- Affected functionality
- Related dependent functionality
- Positive scenarios
- Negative scenarios
- Boundary scenarios
- Cross browser scenarios

Regression testing will be conducted in all relevant areas, based on the nature and impact of the error correction.

## 10. Roles and Responsibilities

| Role                     | Responsibility                              |
|--------------------------|---------------------------------------------|
| QA/Test Engineer         | Test design, execution and defect reporting |
| Developer                | Defect investigation and fixing             |
| Business Analyst         | Requirement clarification                   |
| Product Owner            | Business acceptance and prioritization      |
| UI/UX Team               | UI and layout defect resolution             |
| Backend/Integration Team | Email, SMS, database and API issues         |
| Release/DevOps Team      | Build and environment deployment            |

# 11. Risks and Mitigation

| Risk                                                                          | Impact   | Mitigation                                                                         |
|-------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------|
| Application may become unresponsive when handling invalid or unexpected input | Critical | Apply proper input validation, error handling, and exception handling              |
| Validation rules may behave inconsistently across different modules           | High     | Define and apply consistent validation rules across the application                |
| Date selection or date handling may behave incorrectly                        | Critical | Validate date formats, boundary conditions, date parsing, and date picker behavior |
| Email or SMS integration may not work as expected                             | High     | Perform integration testing using dedicated test accounts and test mobile numbers  |
| Browser specific compatibility issues may occur                               | Medium   | Execute cross browser testing and regression testing on supported browsers         |
| Error messages may be unclear or misleading                                   | Medium   | Verify error messages for accuracy, clarity, and consistency                       |
| Special characters or unexpected input may not be handled correctly           | High     | Perform positive, negative, boundary, and input validation testing                 |
| Some planned test cases may remain unexecuted                                 | Medium   | Track test execution status and monitor No Run and Blocked test cases              |
| Changes or defect fixes may affect existing functionality                     | High     | Perform regression testing on affected and related functionalities                 |
| Test environment or network issues may interrupt testing                      | Medium   | Verify environment availability and resolve environment issues before execution    |

# 12. Test Deliverables

The following deliverables will be prepared:

- Test Plan
- Test Scenario Document
- Test Case Document
- Test Data
- Defect Reports
- Screenshots/Evidence
- Regression Test Results
- Test Execution Summary

- Requirement Traceability Matrix
  - Final Test Summary Report
  - Release Recommendation
- 

## 13. Recommended Smoke Test

For every new build, the following critical flow should be tested first:

**Registration:**

Create Profile → Validate Fields → Submit → Account Created

**Login:**

Login → Dashboard → Logout

**Booking:**

Homepage → Select Route → Date → Passenger → Search Flights

**Flight Status:**

Flight Status → Route/Flight Number → Date → Search

**Profile:**

Login → Update Profile → Change Phone → Save → Verify

**Help:**

Contact Us → Enter Valid Data → Submit → Confirmation

If any critical customer journey fails, the build should be investigated before performing extensive regression testing.